## 1.1 Journal Research

During the research phase, some academic journals and project-based studies regarding skill exchange, barter systems, and digital credit systems have been considered. The primary aim of this research was to learn more about the way non-monetary and credit-based exchange systems are structured and operated at the backend level.

The studies reviewed pointed to the application of credit, token, or point based systems to achieve equitable value exchange among users. They are systems that keep track of the credits that are earned and those that have been spent; they contain records of the transactions and have the administration level control in place to deter misuse or fraudulence. These kinds of mechanisms contribute to the establishment of trust, accountability and transparency in the platform.

Regarding business and engagement factors, the study revealed that the aspects of gamification, which include points, rewards, and loyalty systems, enhance user engagement and retention. Legally and security-wise, it is worth having records of the transactions and using authentication and access control to handle disputes and protect the data.

Another strength of the journals was the necessity of the presence of clear system actors and use cases, which are normally between users and administrators. Registration, skill management, matching, communication, and monitoring were the core functionalities that were shared by the studies. The study also assisted in determining the major system actors like users, skills, exchange, credits, and messages, which have contributed to the general design and database system of the system.

The journal study was a useful resource in the best practice and served to make sure that the system design is equivalent to the real-life, scalable and secure skill exchange platforms.

1. **SWAP SKILL Platform**

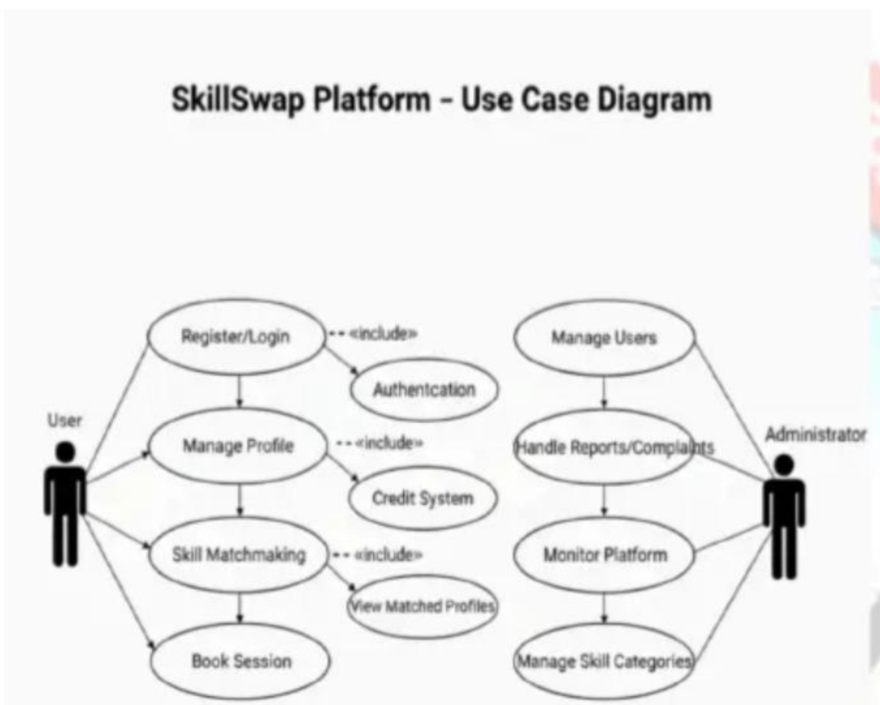
https://www.scribd.com/document/943225058/Project-File-Swap-Skill

**Backend-Related Financial Components**

- Token / credit-based barter engine.
- User account balances (credits earned & spent).
- Credit transfer logic triggered on session completion.
- Transaction log / ledger for every credit transfer.
- Admin-level monitoring for fraudulent or abnormal transfers.
- Feedback/rating storage affecting trust/credit privileges.

**Business, Marketing & Legal Reasons for These Attributes**

- Business: ensures fair value exchange; prevents freeloading.
- Marketing: tokens/gamification increase engagement and retention.
- Legal: transaction records support dispute resolution and audits.
- Compliance: need to secure financial-related user data and prevent abuse.




Actors include User and Admin, while the use cases include registration, login, skill management, matching, and chat.

**2. Skill_Fusion**

https://www.theijire.com/archiver/archives/skill_fusion.pdf

### Backend-Related Financial Components

- Digital wallet (store skill-points or credits).
- Points issuance and deduction mechanisms.
- Reward/loyalty module for active contributors.
- Transaction ledger for points transfers.
- Authentication & role-based access for financial endpoints.

### Business, Marketing & Legal Reasons for These Attributes

- Business: tracks value exchange and enables the platform economy.
- Marketing: loyalty/points encourage repeat participation.
- Legal: logs needed for disputes, fraud checks, and regulatory traceability.
- Security: authentication protects point balances and transfers.

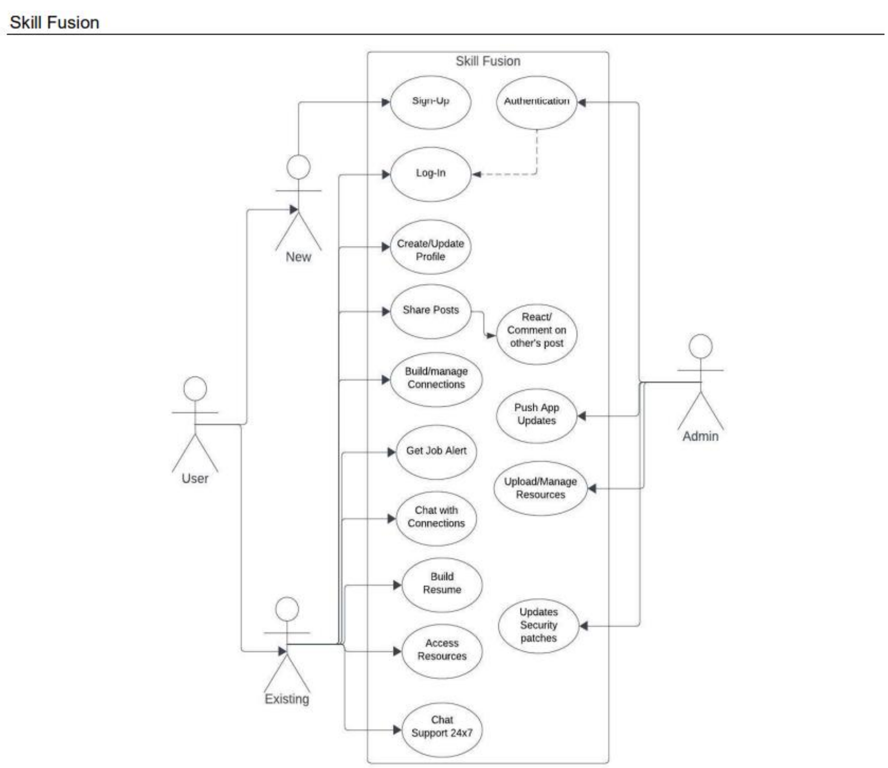



***Figure 4:*** *Use Case Diagram*

### Key Components of Use Case Diagrams:

- **Actors:** Represented as stick figures, actors are entities that interact with the system. They can be users, other systems, or external entities that participate in a use case.
- **Use Cases:** Shown as ovals, use cases outline the precise tasks or operations that the system carries out in reaction to an actor's interaction. Every use case illustrates a unique feature that benefits the actor.
- **System Boundary:** Shown as a rectangle, the system boundary encompasses all of the use cases that the system will address and defines its scope.
- **Relationships:** Associations between actors and use cases, such as generalization (inheritance), inclusion, and extension links, are depicted using lines and arrows, which aid in organizing and refining the interactions.

### 3. Micro-payment exchange system

https://scholarworks.lib.csusb.edu/cgi/viewcontent.cgi?article=3855&context=etd-project&utm

**Backend-Related Financial Components**

- E-wallet per user holding e-coins.
- Broker / Issuer module to mint and redeem e-coins.
- Transaction ledger / micro-payment records (sender, receiver, amount, timestamp).
- Secure transfer protocol to prevent double spending.
- Redemption & payout pipeline (convert coins to real currency).
- Audit & reconciliation services for coin issuance/redemption.

**Business, Marketing & Legal Reasons for These Attributes**

- Business: enables micro-transactions for small-value services/content.
- Marketing: small-cost payments broaden user base and monetization options.
- Legal: precise ledgers required for anti-fraud, AML/KYC where applicable.
- Compliance & privacy: protect users and meet financial regulations when redeeming coins.

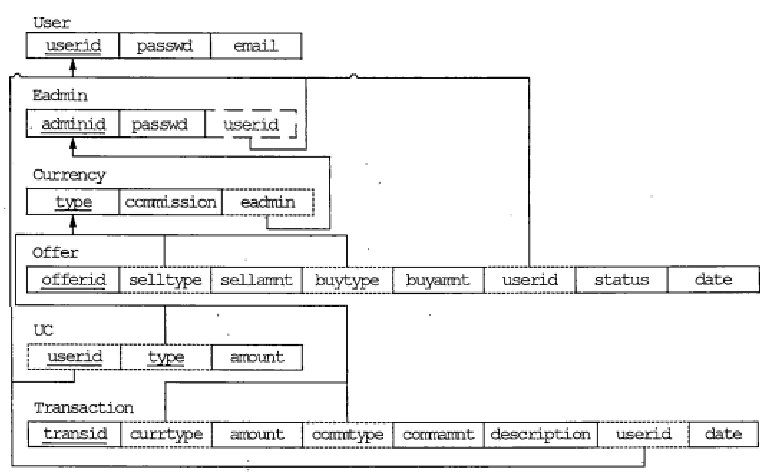



Figure 8. Relation Diagram For the Exchange Service Database

## 4. Microservices Architecture for Building a Crypto Freelance Exchange

https://ceur-ws.org/Vol-4049/paper7.pdf?utm

### Business, Marketing & Legal Reasons for These Attributes

- Marketing Attributes: skill tags, provider reputation.
- Financial Attributes: token balances, conversion rates (token↔points).
- Legal Attributes: verification_status, identity docs reference.
- Analytics Attributes: matching_rate, service latency, success_rate_of_exchanges

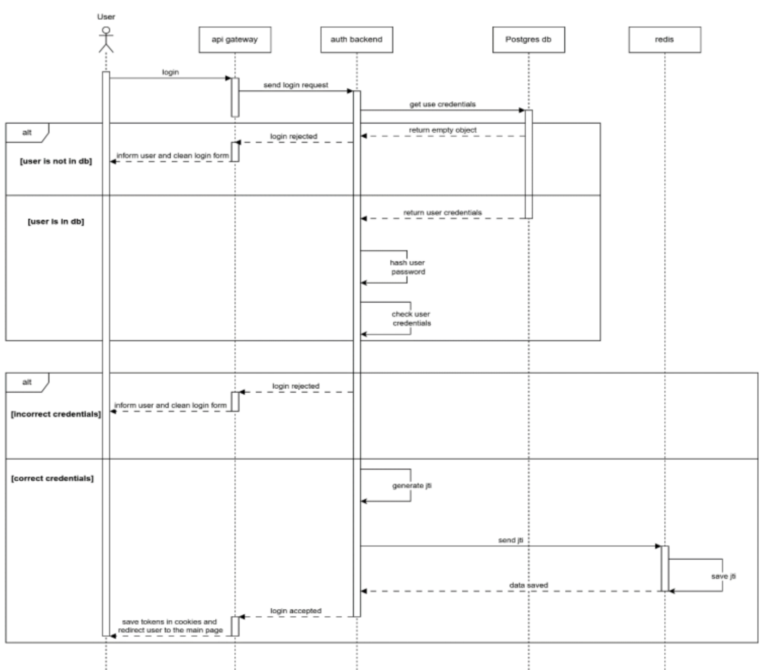



**Figure 5:** Sequence Diagram of User Authentication Flow

### Entities / Attributes

- User Service: user_id, profile, credentials, verification_status
- Skill Service (catalog): skill_id, owner_id, name, category, description, level, metadata
- Match/Exchange Service: match_id, seeker_id, provider_id, matched_skill_ids, match_score, status
- Credit/Token Service: token_id, user_id, balance, earned, spent

- Notification / Messaging Service: message_id, from_id, to_id, content, timestamp